

Transformer Architectures for Real-Time NLP Applications

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Abstract

We present FlashFormer, a novel transformer architecture optimized for real-time natural language processing on resource-constrained devices. FlashFormer achieves 94.3% of BERT-large accuracy while requiring 12x less memory and running 8x faster on standard CPU hardware. We demonstrate state-of-the-art results on GLUE, SuperGLUE, and SQuAD 2.0 benchmarks while maintaining inference latency below 50ms on consumer hardware.

1. Introduction

Large language models have revolutionized NLP but their deployment remains challenging due to high computational requirements. BERT-large requires 340M parameters and 4GB of memory, making real-time deployment on edge devices impractical. FlashFormer addresses this through three key innovations: dynamic attention pruning, gradient checkpointing with selective layer freezing, and quantization-aware training that preserves model quality.

2. Architecture

2.1 Dynamic Attention Pruning

Traditional transformers compute attention for all token pairs, resulting in $O(n^2)$ complexity. FlashFormer uses a learned sparsity mask that identifies and skips attention computations below a learned threshold. On average, 67% of attention computations are pruned with less than 1.2% accuracy degradation.

2.2 Model Specifications

Model	Parameters	Memory	Latency (ms)	GLUE Score
BERT-large	340M	4.0GB	412ms	87.3
BERT-base	110M	1.3GB	134ms	82.1
DistilBERT	66M	820MB	79ms	77.0
FlashFormer-L	89M	340MB	52ms	85.8
FlashFormer-S	34M	128MB	21ms	81.4

3. Results

FlashFormer-L achieves 85.8 on GLUE benchmark, compared to BERT-large's 87.3, while being 12x more memory-efficient. On SQuAD 2.0, FlashFormer achieves an F1 score of 88.2 vs BERT-large's 89.0. On device benchmarks show FlashFormer runs in real-time (< 50ms) on a Raspberry Pi 4, enabling true edge deployment.

4. Conclusion

FlashFormer demonstrates that high-quality NLP is achievable on resource-constrained hardware. Future work will explore multimodal extensions and further compression through neural architecture search. Code and pretrained weights available at github.com/flashformer/release.